Real-Time Rendering: A Practical Introduction

Real-time rendering generates images at interactive frame rates, typically within sixteen milliseconds per frame.
Modern GPUs execute thousands of shader programs in parallel.

A rendering pipeline transforms three-dimensional geometry into a two-dimensional image through a sequence of stages, including vertex processing, rasterization, and shading.

Shadow mapping renders the scene from the light perspective and compares depths to decide whether a point is in shadow.
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Chapter 2: Core Concepts

Latency matters more than raw throughput in interactive applications. Users perceive frames above twenty milliseconds as stutter, so budgets drive every design decision.

Percentage-closer filtering softens the hard shadow edges by averaging multiple depth comparisons per pixel.